

## COMBINED-TECHNIQUE COMPETITION PROBLEM

## Lagrange Interpolation and Coefficient Comparison

Target level: late AIME / introductory olympiad

## Problem

Let  $t$  be a positive integer different from each of 1, 2, 3, 4, 6, so that the six numbers 1, 2, 3, 4, 6,  $t$  are distinct. For each of these six numbers  $v$ , let  $D(v)$  denote the product of the five differences  $(v - u)$  taken over the other five numbers  $u$ . Suppose that

$$\frac{1^7}{D(1)} + \frac{2^7}{D(2)} + \frac{3^7}{D(3)} + \frac{4^7}{D(4)} + \frac{6^7}{D(6)} + \frac{t^7}{D(t)} = 497.$$

Find  $t$ .

**Review note.** The problem appears alone on this page so it can be read without spoilers. A complete solution begins on the next page.

Claude panel: 3/3 answer agreement, well-posedness accepted, difficulty 6.5/10, elegance 5/5.

## Solution

Write the six nodes as

$$\{x_0, x_1, \dots, x_5\} = \{1, 2, 3, 4, 6, t\}, \quad w_i = \frac{1}{\prod_{j \neq i} (x_i - x_j)} = \frac{1}{D(x_i)}.$$

The Lagrange basis polynomial

$$L_i(x) = w_i \prod_{j \neq i} (x - x_j)$$

has degree 5 and leading coefficient  $w_i$ .

### A generating identity

Consider

$$G(z) = \sum_i w_i \prod_{k \neq i} (1 - x_k z),$$

a polynomial of degree at most 5. Since every node is nonzero, evaluate at  $z = 1/x_\ell$ . Every term with  $i \neq \ell$  vanishes, so

$$\begin{aligned} G(1/x_\ell) &= w_\ell \prod_{k \neq \ell} \left(1 - \frac{x_k}{x_\ell}\right) \\ &= w_\ell \frac{\prod_{k \neq \ell} (x_\ell - x_k)}{x_\ell^5} = \frac{1}{x_\ell^5}. \end{aligned}$$

Thus  $G(z) - z^5$  has degree at most 5 and vanishes at the six distinct points  $1/x_i$ . Therefore  $G(z) = z^5$ . Dividing by  $\prod_i (1 - x_i z)$  gives

$$\sum_i \frac{w_i}{1 - x_i z} = \frac{z^5}{\prod_i (1 - x_i z)} = z^5 \sum_{k \geq 0} h_k z^k,$$

where  $h_k$  is the complete homogeneous symmetric polynomial of degree  $k$  in the six nodes.

### Comparing coefficients

The coefficient of  $z^m$  on the left is  $\sum_i w_i x_i^m$ , while the coefficient on the right is  $h_{m-5}$ . In particular, when  $m = 7$ ,

$$\sum_i \frac{x_i^7}{D(x_i)} = \sum_i w_i x_i^7 = h_2.$$

The left side is exactly the sum in the problem.

For six variables,

$$h_2 = \frac{e_1^2 + p_2}{2}.$$

Here

$$e_1 = 1 + 2 + 3 + 4 + 6 + t = 16 + t, \quad p_2 = 1 + 4 + 9 + 16 + 36 + t^2 = 66 + t^2.$$

Consequently,

$$\begin{aligned} h_2 &= \frac{(16 + t)^2 + 66 + t^2}{2} \\ &= t^2 + 16t + 161. \end{aligned}$$

### Solving for the node

Setting  $h_2 = 497$  gives

$$t^2 + 16t - 336 = (t - 12)(t + 28) = 0.$$

The only positive solution is  $t = 12$ , which is distinct from 1, 2, 3, 4, 6.

Therefore, the answer is 12.